MULTIPLYING ONE STRUCTURE MAP OF A TWO-ONE-PERIODIC COMPLEX BY A PARAMETER OF VANISHING SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed. The author proves only the first of the two displayed formulas and says so; this is reproduced as written.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. Same-chapter references are located at their \label lines inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Two-one-periodic complexes and their multiplicity e_R. Same chapter, Definition environment labelled definition-periodic-complex (source line 110) together with the displayed definition of e_R at source lines 200-205 (label equation-multiplicity-coker-ker, line 203). Supplies the objects (M, phi, psi), the quantity e_R and the displayed identity used at proof line 509; the two formulas are the counted content.
- x-power torsion submodule and x-power torsion freeness of the quotient. More on Algebra, Lemma labelled more-algebra-lemma-divide-by-torsion (canonical Stacks tag 05EB); invoked at proof lines 483-484. Supplies that M" = M/M[x^infinity] is x-power torsion free; the reduction to that quotient is retained in the counted span.
- Vanishing criteria for the multiplicities of the summands. Same chapter, Lemma labelled lemma-compare-periodic-lengths (source line 283, Stacks tag 0EA9), invoked at proof lines 494-496; and Lemma labelled lemma-additivity-periodic-length (source line 209, Stacks tag 0EA7), invoked at proof line 497. Supply the three vanishing equalities for the torsion part and the additivity used to discard it; the reduction and the final exact sequence are inside the counted span.

Source reference. Original lemma 05EB: sources/stacks-chow/tags.txt, source lines 6984–6984. Cross-chapter standard prerequisite invoked once at proof lines 483-484 for the x-power torsion freeness of the quotient; the reduction built on it is retained in the counted span.

Source reference. Original lemma 0EA9: sources/stacks-chow/pinned-original-source.tex, source lines 283–283. Same-chapter pointer invoked at proof lines 494-496 for the

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three vanishing equalities on the torsion part; the reduction and the final computation are retained in the counted span.

Source reference. Original lemma 0EA7: sources/stacks-chow/pinned-original-source.tex, source lines 209–209. Same-chapter pointer invoked at proof line 497 for additivity of e_R, which is what allows discarding the torsion part.

Source reference. Original equation 0EA6: sources/stacks-chow/pinned-original-source.tex, source lines 203–203. Same-chapter pointer to the displayed definition of e_R, invoked at proof line 509 to convert the exact sequence into the asserted identity; the exact sequence itself is retained in the counted span.

Lemma 0.1. *Let $(R, \mathfrak{m})$ be a Noetherian local ring. Let (M, φ, ψ) be a $(2,1)$-periodic complex over R with M finite and with cohomology groups of finite length over R. Let $x \in R$ be such that $\dim(\mathit{Supp}(M/xM)) \leq 0$. Then*

$$e_R(M, x\varphi, \psi) = e_R(M, \varphi, \psi) - e_R(\mathrm{Im}(\varphi), 0, x)$$

and

$$e_R(M, \varphi, x\psi) = e_R(M, \varphi, \psi) + e_R(\mathrm{Im}(\psi), 0, x)$$

Proof. We will only prove the first formula as the second is proved in exactly the same manner. Let $M' = M[x^\infty]$ be the x-power torsion submodule of M. Consider the short exact sequence $0 \to M' \to M \to M'' \to 0$. Then M'' is x-power torsion free (More on Algebra, Lemma 05EB). Since φ, ψ map M' into M' we obtain a short exact sequence

$$0 \to (M', \varphi', \psi') \to (M, \varphi, \psi) \to (M'', \varphi'', \psi'') \to 0$$

of $(2,1)$-periodic complexes. Also, we get a short exact sequence $0 \to M' \cap \mathrm{Im}(\varphi) \to \mathrm{Im}(\varphi) \to \mathrm{Im}(\varphi'') \to 0$. We have $e_R(M', \varphi, \psi) = e_R(M', x\varphi, \psi) = e_R(M' \cap \mathrm{Im}(\varphi), 0, x) = 0$ by Lemma 0EA9. By additivity (Lemma 0EA7) we see that it suffices to prove the lemma for (M'', φ'', ψ''). This reduces us to the case discussed in the next paragraph.

Assume $x : M \to M$ is injective. In this case $\mathrm{Ker}(x\varphi) = \mathrm{Ker}(\varphi)$. On the other hand we have a short exact sequence

$$0 \to \mathrm{Im}(\varphi)/x\,\mathrm{Im}(\varphi) \to \mathrm{Ker}(\psi)/\,\mathrm{Im}(x\varphi) \to \mathrm{Ker}(\psi)/\,\mathrm{Im}(\varphi) \to 0$$

This together with (0EA6) proves the formula. □